

NATIONAL TEACHERS COLLEGE
School of Arts, Sciences, and Technology — BSIT Program

NTC_CC105 – DATA STRUCTURE W/LAB

WEEK 5 LABORATORY INSTRUCTION

Laboratory Activity 5

Instructor	Prof. Darwin C. Vargas, MIT
Program	BSIT
Academic Year	2026–2027
Programming Language	C++

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3. Section	2.4 BSIT
4. Date	Oct 5, 2026
5. Laboratory Activity No.	Laboratory Activity 5
6. Laboratory Title	Linear Architecture Synthesis & Peer Code Audit

Part C: Big-O Algorithmic Complexity Analysis

Operation	Target Code / Function	Best Case	Average Case	Worst Case	Space Complexity	Mathematical Justification
Unamortized Insert	add_item()	$O(1)$	$O(1)$	$O(n)$	$O(n)$	$O(1)$ when capacity remains. Worst case $O(n)$ occurs when array reaches capacity, requiring reallocation and copying n elements.
Linear Deletion	remove_item_at()	$O(1)$	$O(n)$	$O(n)$	$O(1)$	Deleting the last element takes $O(1)$. Deleting at index 0 requires shifting $n-1$ elements left. The legacy double loop does not change asymptotic complexity, but adds constant factor overhead.
Linear Search	findItem()	$O(1)$	$O(n)$	$O(n)$	$O(1)$	Best case $O(1)$ occurs if the target is at index 0. Worst case $O(n)$ occurs if target is at index $n-1$ or absent.
Array Display	printAll()	$O(n)$	$O(n)$	$O(n)$	$O(1)$	Must sequentially visit and display every element from index 0 to $n-1$.
Matrix Transpose	processMatrix()	$O(R \times C)$	$O(R \times C)$	$O(R \times C)$	$O(R \times C)$	Requires visiting every cell in an $R \times C$ grid via nested loops.